

“Having a time correlated view of analogue, digital and RF signals on a single display has never been available before”



Preston Gabel, senior marketing manager, Tektronix, while talking to Atul Goel from EFY magazine, about some new trends in T&M, reveals about the company's new MDO4000 series that combines an oscilloscope and a spectrum analyser in a single instrument

■ **PRESTON GABEL**
SENIOR MARKETING MANAGER, TEKTRONIX

What are the top five tech-trends in oscilloscopes?

Oscilloscope has been the tool of choice for design engineers for over 65 years. Over that long history, oscilloscopes have become faster, and have been adapted to address a broader set of design and debug challenges. Modern oscilloscopes offer more value than ever before.

More recent history suggests that we'll continue to see the proliferation of designs that incorporate high-speed and efficient low-speed serial standards for transport of data within and between systems. Many new designs have stringent power constraints resulting in unique design and measurement challenges for engineers.

Properly handling often 'delicate' signals from the device-under-test to the oscilloscope display has always been a challenge and we expect that trend will continue.

With the world of analogue and digital getting more and more integrated at the design engineering level, what are the kind of demands being made by your customers?

To be specific, on the digital side, we regularly receive requests for support for new serial standards or support for variations on already supported standards. Customers have also become more aware and vocal about how test equipment impacts their sensitive analogue and digital circuits.

What is the USP of MDO4000 series? How do you think it will change the way people work with oscilloscopes?

MDO4000 is the first and only product to combine an oscilloscope and a spectrum analyser in a single instrument. You would expect that the resulting product would do what each individual product does, but the combination yields so much more! Having a time correlated view

of analogue, digital and radio frequency (RF) signals on a single display has never been available before. Tektronix based the MDO4000 series upon the world's most popular mixed-signal oscilloscope. This results in an instrument with unique RF capabilities yet it is surprisingly familiar to operate. The oscilloscope is generally the most used piece of test gear on the engineer's bench. The addition of RF capability in MDO4000 extends the usefulness of the engineer's tool of choice to RF and the frequency domain.

How will it benefit design engineers?

Today, embedded system design engineers are facing extremely difficult design challenges. Furthermore, project budgets are tight and timelines are short. The added complexity of wireless connectivity required by many of today's products has made their job more difficult than ever before. MDO4000 brings critical capabilities together in one instrument to solve these complex problems. Set-ups (and data collection) that took hours, days and in some cases even weeks, can now be accomplished in seconds. MDO4000 will help engineers identify issues quickly so they can get to solutions.

Is a combined product really needed when users already have two separate products with similar capabilities?

Yes. With MDO4000 you get the functionality of a scope and a spectrum analyser in a single instrument. But more importantly you get a time correlated view of analogue, digital and RF signals on a single display. This type of view has never been available before and is incredibly useful for debugging modern embedded designs incorporating wireless connectivity. ●